

Contact

tarunisco@gmail.com

www.linkedin.com/in/tarunisco
(LinkedIn)

Top Skills

Multi-agent Systems

Retrieval-Augmented Generation
(RAG)

Technical Solution Design

Certifications

Machine Learning Advanced
Nanodegree

Tarun Bhardwaj

Senior Data Scientist @ Fractal | Building GenAI solutions
Gurugram, Haryana, India

Summary

I'm a Senior Data Scientist and Generative AI Solution Architect passionate about building intelligent, scalable systems that turn data into actionable insights. With over six years of experience in machine learning, data engineering, and GenAI applications, I specialize in designing agentic AI platforms, LLM-powered automation, and insight-generation systems that drive measurable business value. My work focuses on creating end-to-end AI solutions — from Text-to-SQL and RAG architectures to agentic workflows and multimodal intelligence — enabling seamless collaboration between humans and AI. I've worked extensively with tools and frameworks such as Python, Pyspark, Azure, Databricks, FastAPI, LangChain, LangGraph, OpenAI Agents SDK, Google ADK, and MCP, building robust and reusable components for enterprise-scale AI platforms. Beyond technology, I enjoy mentoring teams, shaping GenAI strategy, and exploring new ways to make AI more intuitive, explainable, and impactful across organizations.

Experience

Fractal

4 years 5 months

Senior Data Scientist

April 2025 - Present (1 year 1 month)

Gurugram

- Leading the end-to-end architecture, design, and development of a centralized GenAI platform supporting multiple Commercial IT use cases for a global healthcare client. Each use case features an agentic or multi-agent system delivering insights, resolutions, or workflow automations. The platform's backbone consists of scalable agentic Text-to-SQL and RAG pipelines, designed for easy customization across use cases. All platform components are modular and accessible via an MCP server, enabling seamless integration and standardized access for building agentic workflows. Additionally, the framework allows cross-domain collaboration, enabling teams

outside Commercial IT to leverage domain intelligence without requiring deep subject expertise.

- Driving the solution design and development of an agentic Text-to-SQL system tailored for the Personal Healthcare Equipment domain, facilitating natural language–driven insight generation. Collaborating with a parallel project team as a GenAI subject matter expert, providing architectural guidance and implementation support to optimize performance and contextual accuracy.
- Providing technical mentorship to multiple project teams across the organization, guiding them in designing and developing GenAI applications focused on insight generation, intelligent automation, and agentic workflow orchestration.
- Leading internal GenAI capability development at Fractal by building state-of-the-art frameworks, reusable components, and solution architectures that can be leveraged for client proposals or integrated into existing projects to enhance scalability, adaptability, and intelligence.
- Actively contributing to client engagement and innovation by conceptualizing and developing proposals, proofs of concept (POCs), and new GenAI use cases, aligning cutting-edge AI capabilities with evolving business needs to drive measurable impact and digital transformation.

Data Scientist

December 2021 - March 2025 (3 years 4 months)

Gurugram

- Developed a customized Text-to-SQL framework using LangGraph to empower sales and business teams with on-demand, conversational insights from structured enterprise data. Leveraged generative AI and LLM-based query understanding to translate natural language questions into optimized SQL queries, enabling non-technical users to seamlessly interact with databases, uncover trends, and make data-driven decisions without technical dependencies.
- Developed a scalable framework for the Customer Lifetime Value (CLV) program by integrating data from multiple enterprise systems to generate high-quality upsell opportunities for installed medical equipment. Supported the

creation of the CLV Playground tool on CRM Analytics, contributing to a new sales funnel valued at approximately €6 million per quarter.

- Delivered a comprehensive NMPA Licenses connected data report, providing critical visibility into €19 million worth of open sales opportunities at risk in the China market, enhancing decision-making and sales pipeline management.
- Designed a 3D analytical framework to optimize trade-in strategies for professional medical equipment, incorporating equipment health indices, customer segmentation, and deal performance analytics to maximize revenue and margins. Delivered a proof of concept identifying ultrasound devices suitable for trade-in across Australia and New Zealand.

Cognam Technologies Pvt. Ltd.

3 years

Machine Learning Team Lead

August 2021 - December 2021 (5 months)

New Delhi

Led a team of five in developing an identity resolution system for transactional data to accurately segment customers and enhance data integrity. Researched and guided a project leveraging image captioning and reinforcement learning to automate the conversion of UI mockups into HTML/CSS structures, integrating computer vision and NLP techniques for intelligent front-end generation.

Machine Learning Engineer

May 2019 - July 2021 (2 years 3 months)

New Delhi, Delhi, India

Developed and deployed a Python Flask-based end-to-end face recognition system to optimize authentication workflows and reduce agent login time. Built a spoof detection model using image distortion analysis to prevent photo and video-based attacks. Designed and deployed an ensemble machine learning model for fraud transaction detection, incorporating automated retraining, feature engineering, and performance monitoring to ensure high model accuracy and robustness.

Software Engineer Trainee

January 2019 - May 2019 (5 months)

New Delhi Area, India

Conducted data analysis on financial transaction datasets to identify fraud patterns and enhance payment security, applying feature engineering and

machine learning techniques using classifiers such as logistic regression, decision trees, and random forests to build predictive fraud detection models.

CampusHaat Solutions

Android Developer Intern

May 2018 - July 2018 (3 months)

IIT Kanpur

Developed and integrated key user interaction features, including real-time chat and discussion portals, into an Android application using Firebase to enhance communication and engagement within the platform.

Education

Indian Institute of Information Technology Vadodara

Bachelor of Technology - BTech, Information Technology · (2015 - 2019)

Shriji Baba Saraswati Vidya Mandir

Class 6 - Class 12, Intermediate, Science · (2008 - 2015)